

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

CS2 - Risk Modelling and Survival Analysis

(Paper B)

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1)

```
##### Q1

### i.

## Defining transition rates (mu) values

muMD <- 0.012
muKD <- 0.032
muBD <- 0.026
muDM <- 0.02
muKM <- 0.029
muBM <- 0.014
muDK <- 0.022
muMK <- 0.002
muBK <- 0.025
muDB <- 0.009
muMB <- 0.041
muKB <- 0.036

## Defining retained rates
muDD <- -muDM-muDK-muDB
muMM <- -muMD-muMK-muMB
muKK <- -muKD-muKM-muKB
muBB <- -muBD-muBM-muBK

## Arranging transition rates and storing in an array
transition.rates <- c( muDD, muMD, muKD, muBD, muDM, muMM, muKM, muBM,
muDK, muMK, muKK, muBK, muDB, muMB, muKB, muBB)

## Converting the array into a matrix of 4 dimension
A <- matrix(transition.rates,4,4)

## Setting dimnames for the matrix
teams <- c("D", "M", "K", "B")
dimnames(A) <- list(teams,teams)

## Printing the matrix
A
```

	D	M	K	B
D	-0.051	0.020	0.022	0.009
M	0.012	-0.055	0.002	0.041
K	0.032	0.029	-0.097	0.036
B	0.026	0.014	0.025	-0.065

(3)

ii.

Over a small period of time t , the transition probability is given by

$$P(t) = I + h * A + O(t)$$

Where,

I is identity matrix,

O(t) is higher order polynomial matrix of t

To approximate, O(t) is ignored

t <- 1/12

P12 = diag(4) + t*A

Printing the matrix

P12

	D	M	K	B
D	0.995750000	0.001666667	0.001833333	0.000750000
M	0.001000000	0.995416667	0.000166667	0.003416667
K	0.002666667	0.002416667	0.991916667	0.003000000
B	0.002166667	0.001166667	0.002083333	0.994583333

t <- 1/365

P365 = diag(4) + t*A

Printing the matrix

P365

	D	M	K	B
D	9.998603e-01	5.479452e-05	6.027397e-05	2.465753e-05
M	3.287671e-05	9.998493e-01	5.479452e-06	1.123288e-04
K	8.767123e-05	7.945205e-05	9.997342e-01	9.863014e-05
B	7.123288e-05	3.835616e-05	6.849315e-05	9.998219e-01

(4)

iii.

$P(18 \text{ years}) = P(1 \text{ year})^{18} = (P(1/12 \text{ year})^{12})^{18} = (P(1/12 \text{ year}))^{(12*18)} = (P(1/12 \text{ year}))^{216}$

Finding P18y using P12

time = 18*12

temp = diag(4)

for (j in 1:time){

temp = temp %*% P12

}

P18y1 = temp

P18y1

	D	M	K	B
[1,]	0.4895653	0.2016751	0.14311265	0.1656469
[2,]	0.1736954	0.4467124	0.08281176	0.2967805
[3,]	0.2559005	0.2249033	0.25752055	0.2616757

```
[4,] 0.2447564 0.1746455 0.15667677 0.4239213
```

```
## Finding P18y2 using P365
```

```
time = 18*365
```

```
temp = diag(4)
```

```
for (j in 1:time){
```

```
  temp = temp %*% P365
```

```
}
```

```
P18y2 = temp
```

```
P18y2
```

```
      D      M      K      B
[1,] 0.4902865 0.2013707 0.14292641 0.1654164
[2,] 0.1734771 0.4475194 0.08267705 0.2963264
[3,] 0.2555416 0.2245916 0.25855080 0.2613160
[4,] 0.2443847 0.1743930 0.15646440 0.4247579
```

```
P18y1-P18y2
```

```
      D      M      K      B
[1,] -0.0007211687 0.0003043610 0.0001862399 0.0002305679
[2,] 0.0002182229 -0.0008070297 0.0001347096 0.0004540973
[3,] 0.0003589330 0.0003116419 -0.0010302422 0.0003596674
[4,] 0.0003717544 0.0002525069 0.0002123678 -0.0008366290
```

Since both are approximations at different level of accuracy, there is a difference among them. The difference is because of the $O(t)$, the error term. The order of magnitude of differences is $10e-4$.

(5)

```
### iv.
```

```
## 10 years
```

```
time = 10*12
```

```
temp = diag(4)
```

```
for (j in 1:time){
```

```
  temp = temp %*% P12
```

```
}
```

```
P10y = temp
```

```
P10y
```

```
      D      M      K      B
[1,] 0.6382293 0.1437287 0.11902439 0.09901768
[2,] 0.1086005 0.6070322 0.04463877 0.23972852
[3,] 0.1966992 0.1747561 0.41995202 0.20859268
[4,] 0.1787322 0.1158165 0.13301133 0.57243992
```

a. Probability that a Delhi fan would turn into Bangalore fan in 10 years is 9.9%

```
## 15 years
```

```
time = 15*12
```

```
temp = diag(4)
for (j in 1:time){
  temp = temp %*% P12
}
P15y = temp
P15y
```

```
      D      M      K      B
[1,] 0.5354921 0.1840814 0.13743990 0.1429866
[2,] 0.1516703 0.4958618 0.06966218 0.2828057
[3,] 0.2392805 0.2110437 0.30194965 0.2477261
[4,] 0.2252365 0.1556543 0.15169813 0.4674111
```

b. Probability that a Mumbai fan would turn into Kolkata fan in 15 years is 6.96%

```
## 4.5 years
time = 4.5*12
temp = diag(4)
for (j in 1:time){
  temp = temp %*% P12
}
P4.5y = temp
P4.5y
```

```
      D      M      K      B
[1,] 0.80422741 0.07752999 0.07413986 0.04410274
[2,] 0.05212199 0.78777722 0.01621477 0.14388602
[3,] 0.11453710 0.10279536 0.65736853 0.12529901
[4,] 0.09858343 0.05832005 0.08375495 0.75934157
```

c. Probability that a Bangalore fan will be a Bangalore fan in 4.5 years is 75.93%

```
time = 4.5*12
Pk4.5y = (P12[3,3])^time
Pk4.5y
```

```
[1] 0.6451491
```

d. Probability that a Kolkata fan will remain Kolkata fan for 4.5 years is 64.51%

```
init <- matrix(c(100,100,100,100),1,4) ## any number can be used in place of 100
init %*% P10y
```

```
      D      M      K      B
[1,] 112.2261 104.1333 71.66265 111.9779
```

e. If each team starts with same number of fans, Delhi would be most popular after 10 years

(Max 10)

```
### v.
```

```
## Setting up the variables
initial <- matrix(c(1000,1000,1000,1000),1,4)
```

```

Sim1 <- as.data.frame(initial)
names(Sim1) <- c("D","M","K","B")
temp <- initial

## simulation using for loop and storing in Sim1
time <- 30*12
for (i in 1:time) {
  temp <- temp %*% P12
  Sim1[i+1,] <- as.vector(temp)
}

## Adding time column
Sim1["Time"] <- 0:time
Sim1 <- Sim1[c("Time","D","M","K","B")] ## Arranging the columns - not necessary
head(Sim1)

```

```

  Time    D      M      K      B
1  0 1000.000 1000.000 1000.0000 1000.000
2  1 1001.583 1000.667  996.0000 1001.750
3  2 1003.154 1001.325  992.0390 1003.482
4  3 1004.711 1001.976  988.1166 1005.196
5  4 1006.256 1002.619  984.2325 1006.893
6  5 1007.788 1003.254  980.3862 1008.572

```

(3)

```

### vi.

```

```

## Entering matrix values
M <- matrix(data = c(99575, 167, 183, 75, 100, 99542, 17, 342, 181, 240, 98282, 297, 217,
117, 208, 99458),4,4,byrow = TRUE)

```

```

P12B <- M
## Dividing each row by row sum to get probabilities
r <- rowSums(M)
for (i in 1:4) {
  P12B[i,] = (1/r[i])*M[i,]
}
dimnames(P12B) <- list(teams,teams)
P12B

```

```

      D      M      K      B
D 0.995750000 0.001670000 0.001830000 0.000750000
M 0.000999990 0.995410046 0.0001699983 0.003419966
K 0.001828283 0.002424242 0.9927474747 0.003000000
B 0.002170000 0.001170000 0.0020800000 0.994580000

```

```

## Setting up the variables
Sim2 <- as.data.frame(initial)
names(Sim2) <- c("D","M","K","B")

```

```
temp <- initial

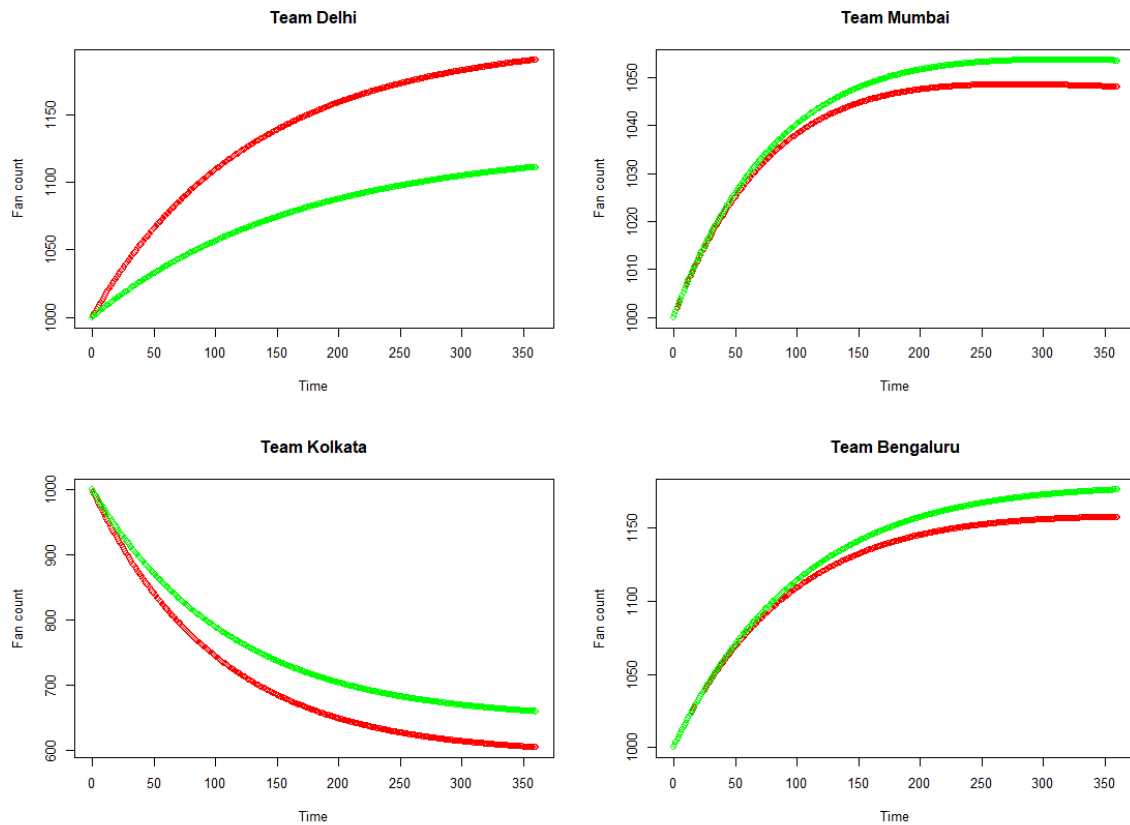
## simulation using for loop and storing in Sim2
for (i in 1:time) {
  temp <- temp %*% P12B
  Sim2[i+1,] <- as.vector(temp)
}

## Adding time column
Sim2["Time"] <- 0:time
Sim2 <- Sim2[c("Time", "D", "M", "K", "B")] ## Arranging the columns - not necessary
head(Sim2)
```

```
Time    D    M    K    B
1  0 1000.000 1000.000 1000.0000 1000.000
2  1 1000.748 1000.674  996.8275 1001.750
3  2 1001.492 1001.341  993.6831 1003.484
4  3 1002.231 1002.000  990.5666 1005.202
5  4 1002.966 1002.653  987.4777 1006.904
6  5 1003.697 1003.297  984.4162 1008.590
```

```
## adding indicator and binding
Sim1["Sim"] <- "red"
Sim2["Sim"] <- "green"
Sim <- rbind(Sim1, Sim2)
head(Sim)
Time    D    M    K    B Sim
1  0 1000.000 1000.000 1000.0000 1000.000 red
2  1 1001.583 1000.667  996.0000 1001.750 red
3  2 1003.154 1001.325  992.0390 1003.482 red
4  3 1004.711 1001.976  988.1166 1005.196 red
5  4 1006.256 1002.619  984.2325 1006.893 red
6  5 1007.788 1003.254  980.3862 1008.572 red
```

```
par(mfrow = c(2, 2))
plot(Sim$Time, Sim$D, col = Sim$Sim, main = "Team Delhi", xlab = "Time", ylab = "Fan count")
plot(Sim$Time, Sim$M, col = Sim$Sim, main = "Team Mumbai", xlab = "Time", ylab = "Fan count")
plot(Sim$Time, Sim$K, col = Sim$Sim, main = "Team Kolkata", xlab = "Time", ylab = "Fan count")
plot(Sim$Time, Sim$B, col = Sim$Sim, main = "Team Bengaluru", xlab = "Time", ylab = "Fan count")
```



From the graphs, we see that the variation between the simulations is high for D and K
indicating the rates of transition from D to K or K to D is the incorrect one.

(6)

vii.

Inversing the formula $P(t) = I + t*B + o(h)$ where $p(t)$ is transition matrix, $o(t)$ is error term, and t is a small interval in time

for a month, $t = 1/12$. To approximate, $o(t)$ is ignored

we have $B = 12*(P(t)-I)$

$B = \text{round}(12*(P_{12B} - \text{diag}(4)), 3)$

B-A

	D	M	K	B
D	0.00	0	0.00	0
M	0.00	0	0.00	0
K	-0.01	0	0.01	0
B	0.00	0	0.00	0

From the matrix it is clear that the error is in transition rate from K to D. It is 0.022 instead of 0.032.

This matches the deduction from previous run

(2)

[33 Marks]**Q. 2)**

```

#### installing packages (as required)
#install.packages('dplyr')
#install.packages('ggplot2')
#install.packages('lubridate')

#### loading Packages
library(dplyr)
library(ggplot2)
library(lubridate)

#### i.
# Load dataset
data <- read.csv("Policy_data.csv", header = TRUE)
data <- data %>%
  mutate(Date.of.birth = as.Date(Date.of.birth, format = "%d-%m-%Y"),
         Date.of.death = as.Date(Date.of.death, format = "%d-%m-%Y"),
         Policy.start.date = as.Date(Policy.start.date, format = "%d-%m-%Y"),
         Policy.end.date = as.Date(Policy.end.date, format = "%d-%m-%Y"))
head(data)

```

Life.num	Gender	Date.of.birth	Policy.start.date	Date.of.death	Policy.end.date
1	1	M	1964-03-23	2020-10-30	<NA> 2023-08-20
2	2	M	1994-02-02	2024-11-23	<NA> 2025-11-23
3	3	M	1969-10-16	2007-08-14	<NA> 2018-10-20
4	4	M	2015-02-24	2024-12-31	<NA> 2025-12-31
5	5	M	1961-03-12	2022-10-05	<NA> 2024-09-18
6	6	M	2004-07-05	2023-05-12	<NA> 2024-09-01

(2)

```

#### ii.
summary(data)

```

Life.num	Gender	Date.of.birth	Policy.start.date
Min. :	1	Length:200000	Min. :1960-01-01
1st Qu.:	50001	Class :character	1st Qu.:1975-04-09
Median :	100001	Mode :character	Median :1990-07-01
Mean :	100001		Mean :2020-10-10
3rd Qu.:	150000		Mean :2018-12-01
Max. :	200000		3rd Qu.:2005-10-23
			3rd Qu.:2024-10-26
			Max. :2020-12-31
			Max. :2024-12-31

Date.of.death	Policy.end.date
Min. :	2004-03-23
1st Qu.:	2004-03-23
Median :	2017-06-20
Mean :	2020-11-27
3rd Qu.:	2020-12-15
Max. :	2024-01-24
NA's :	2019-11-24
	Mean :2022-08-06
	3rd Qu.:2023-04-27
	3rd Qu.:2025-10-26
	Max. :2024-12-30
	Max. :2025-12-31
	NA's :196204

```
head(data %>% filter(Policy.start.date > Policy.end.date),10)
```

	Life.num	Gender	Date.of.birth	Policy.start.date	Date.of.death	Policy.end.date
1	379	M	1964-05-17	2024-05-28	2024-05-02	2024-05-02
2	2432	M	1966-01-21	2024-09-27	2024-01-07	2024-01-07
3	3148	M	1963-10-12	2014-06-18	2013-09-29	2013-09-29
4	3178	M	1975-10-20	2022-12-09	2022-10-08	2022-10-08
5	4738	M	1989-04-22	2024-05-04	2024-04-13	2024-04-13
6	4818	M	2000-02-08	2018-02-19	2018-02-03	2018-02-03
7	5114	M	1966-07-25	2017-05-30	2016-07-12	2016-07-12
8	6524	M	1987-08-05	2019-12-04	2019-07-28	2019-07-28
9	7678	M	1991-12-11	2022-06-11	2021-12-03	2021-12-03
10	9648	M	1961-06-07	2009-11-03	2009-05-26	2009-05-26

```
colSums(is.na(data))
```

Life.num	Gender	Date.of.birth	Policy.start.date
0	0	0	0

Date.of.death	Policy.end.date
196204	0

```
data %>% filter(Policy.start.date < Date.of.birth)
```

```
[1] Life.num      Gender      Date.of.birth  Policy.start.date
[5] Date.of.death  Policy.end.date
<0 rows> (or 0-length row.names)
```

(Max 3)

```
### iii.
```

```
data <- data %>%
  mutate(Policy.start.date = if_else(Policy.start.date > Policy.end.date,Policy.start.date-
years(1),Policy.start.date))
head(data %>% filter(Policy.start.date > Policy.end.date),10)
```

```
[1] Life.num      Gender      Date.of.birth  Policy.start.date Date.of.death
[6] Policy.end.date
<0 rows> (or 0-length row.names)
```

(2)

```
### iv.
```

```
Study.start.date <- as.Date("2008/01/01")
Study.end.date <- as.Date("2023/12/31")
data <- data %>%
  mutate(Exposure.start.date =
    pmax(Policy.start.date, Date.of.birth, Study.start.date),
    Exposure.end.date =
    if_else(is.na(Date.of.death),
      pmin(Policy.end.date, Study.end.date),
      pmin(Date.of.death, Study.end.date)),
    Age.at.start = as.integer(floor(as.numeric(
```

```

      Exposure.start.date - Date.of.birth) / 365.25)),
      Age.at.end = as.integer(floor(as.numeric(
      Exposure.end.date - Date.of.birth) / 365.25)),
      Age.at.death = as.integer(floor(as.numeric(
      Date.of.death - Date.of.birth) / 365.25))) %>%
      filter(Exposure.start.date<=Exposure.end.date)
head(data)

```

	Life.num	Gender	Date.of.birth	Policy.start.date	Date.of.death	Policy.end.date
1	1	M	1964-03-23	2020-10-30	<NA>	2023-08-20
2	3	M	1969-10-16	2007-08-14	<NA>	2018-10-20
3	5	M	1961-03-12	2022-10-05	<NA>	2024-09-18
4	6	M	2004-07-05	2023-05-12	<NA>	2024-09-01
5	7	M	2000-11-25	2021-01-19	<NA>	2022-10-27
6	8	M	1974-11-24	2012-04-05	2016-11-13	2016-11-13

	Exposure.start.date	Exposure.end.date	Age.at.start	Age.at.end	Age.at.death
1	2020-10-30	2023-08-20	56	59	NA
2	2008-01-01	2018-10-20	38	49	NA
3	2022-10-05	2023-12-31	61	62	NA
4	2023-05-12	2023-12-31	18	19	NA
5	2021-01-19	2022-10-27	20	21	NA
6	2012-04-05	2016-11-13	37	41	41

(6)

```
### v.
```

```

# Filter rows with valid death age
deaths <- data %>% filter(Age.at.death >= 0)

```

```

# Count deaths by age and gender
death_counts <- deaths %>%
  group_by(Age.at.death, Gender) %>%
  summarise(Deaths = n(), .groups = "drop")

```

```

# Calculate exposures by age and gender
ages <- 0:70
genders <- c("M", "F")
exposures <- expand.grid(Age = ages, Gender = genders) %>%
  rowwise() %>%
  mutate(Exposures = sum(data$Gender == Gender & data$Age.at.start < Age &
data$Age.at.end > Age)) %>%
  mutate(Exposures = Exposures + sum(data$Gender == Gender & (data$Age.at.start ==
Age | data$Age.at.end == Age))/2) %>%
  ungroup()

```

```

# Merge death counts with exposures
mortality_table <- exposures %>%
  left_join(death_counts, by = c("Age" = "Age.at.death", "Gender")) %>%
  mutate(
    Deaths = if_else(is.na(Deaths), 0, Deaths),

```

```

Mortality.rate = Deaths / Exposures
)
summary(mortality_table)

  Age      Gender      Exposures      Deaths
Min. : 0.00  Length:142      Min. : 0      Min. : 0.00
1st Qu.:17.25 Class :character 1st Qu.: 0      1st Qu.: 0.00
Median :35.00 Mode  :character Median :5002      Median : 10.50
Mean  :35.00                      Mean  :4451      Mean  : 26.26
3rd Qu.:52.75                      3rd Qu.:8433      3rd Qu.: 46.50
Max.  :70.00                      Max.  :9006      Max.  :103.00

```

```

Mortality.rate
Min. :0.00000
1st Qu.:0.00134
Median :0.00479
Mean  : Inf
3rd Qu.:0.01766
Max.  : Inf
NA's  :44

```

(7)

```

### vi.
# Creating Life table
life_table <- mortality_table %>%
  filter(Age >=20 & Age <= 50) %>%
  select(Age,Gender, Mortality_rate = Mortality.rate)

# View first few rows of comparison
head(life_table, 10)

```

```

# A tibble: 10 × 3
  Age Gender Mortality_rate
<int> <chr>      <dbl>
1   20 M      0.000306
2   21 M      0.000532
3   22 M      0.000979
4   23 M      0.000582
5   24 M      0.00113
6   25 M      0.000336
7   26 M      0.000779
8   27 M      0.00134
9   28 M      0.000900
10  29 M      0.00180

```

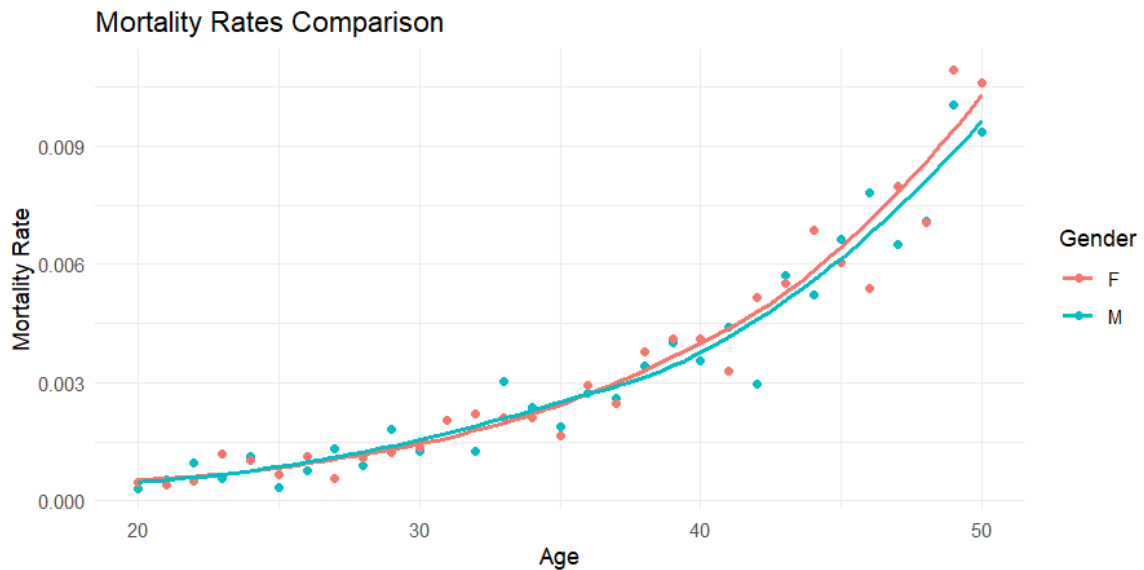
(2)

```

### vii.
ggplot(life_table, aes(x = Age, y = Mortality_rate, colour = Gender))+
  geom_point()+
  geom_smooth(se = FALSE)+

```

```
labs(title = "Mortality Rates Comparison",
     x = "Age", y = "Mortality Rate")+
theme_minimal()
```



From the graph it is clear the mortality rates for both gender are not heavily different. For the ages 25 to 36, males have higher mortality than females. For the ages 37 and above females have higher mortality than males.

(3)

```
### viii.
```

```
## a.
```

```
male_life_table <- filter(life_table, Gender == 'M')
```

```
female_life_table <- filter(life_table, Gender == 'F')
```

```
## b.
```

```
male_gompertz <- lm(log(male_life_table$Mortality_rate) ~ male_life_table$Age)
```

```
female_gompertz <- lm(log(female_life_table$Mortality_rate) ~ female_life_table$Age)
```

```
male_gompertz
```

```
Call:
```

```
lm(formula = log(male_life_table$Mortality_rate) ~ male_life_table$Age)
```

```
Coefficients:
```

```
(Intercept) male_life_table$Age
-9.6453      0.1014
```

```
female_gompertz
```

```
Call:
```

```
lm(formula = log(female_life_table$Mortality_rate) ~ female_life_table$Age)
```

```
Coefficients:
```

```
(Intercept) female_life_table$Age
-9.6371      0.1019
```

```
B_male = exp(as.numeric(coef(male_gompertz)))[1]
```

```
B_male
```

```
[1] 6.472923e-05
```

```
c_male=exp(as.numeric(coef(male_gompertz)))[2]
```

```
c_male
```

```
[1] 1.106719
```

```
B_female=exp(as.numeric(coef(female_gompertz)))[1]
```

```
B_female
```

```
[1] 6.525943e-05
```

```
c_female=exp(as.numeric(coef(female_gompertz)))[2]
```

```
c_female
```

```
[1] 1.107289
```

```
## c.
```

```
male_life_table <- male_life_table %>%
```

```
  mutate(Graduated_rate = round(B_male*c_male^Age,10))
```

```
female_life_table <- female_life_table %>%
```

```
  mutate(Graduated_rate = round(B_female*c_female^Age,10))
```

```
## d.
```

```
life_table <- rbind(male_life_table,female_life_table)
```

```
head(life_table)
```

```
# A tibble: 6 × 4
```

	Age	Gender	Mortality_rate	Graduated_rate
	<int>	<chr>	<dbl>	<dbl>
1	20	M	0.000306	0.000492
2	21	M	0.000532	0.000544
3	22	M	0.000979	0.000602
4	23	M	0.000582	0.000667
5	24	M	0.00113	0.000738
6	25	M	0.000336	0.000817

(6)

```
### ix.
```

```
ggplot(life_table,aes(x = Age, y = Mortality_rate, colour = Gender))+
```

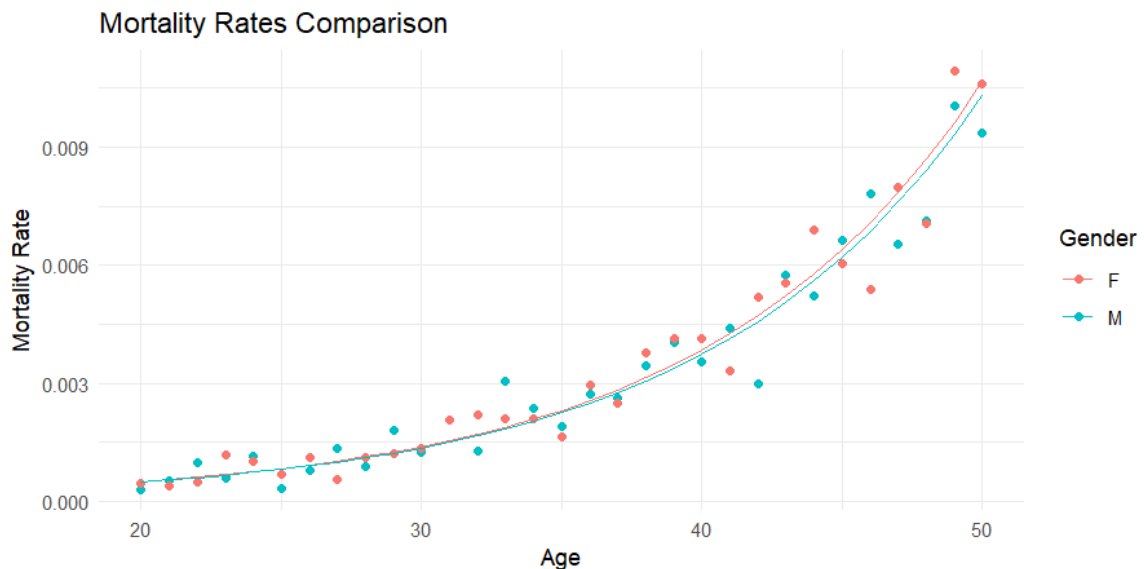
```
  geom_point()+
```

```
  geom_line(aes(y=Graduated_rate))+
```

```
  labs(title = "Mortality Rates Comparison",
```

```
        x = "Age", y = "Mortality Rate")+
```

```
  theme_minimal()
```



#From the graph it is clear the Gompertz fit for mortality rates of both gender are not significantly different.

(3)

x.

```
life_table <- life_table %>%
  mutate(deviations = Mortality_rate-Graduated_rate)
males <- life_table %>% filter(Gender=="M")
females <- life_table %>% filter(Gender=="F")
```

Signs test:

If there are 'm' observations, then half of the deviations ($m/2$ if even (or) $(m+1)/2$ if odd) should be negative and the other half positive. This is a two-tailed test looking for both positive and negative bias. (Consider at 5% significance level)

Null Hypothesis: the number of negative deviations follows Binomial($m, 0.5$)

Alternative Hypothesis: the number of negative deviations doesn't follow Binomial($m, 0.5$)

For Males

```
dev_neg <- sum(males$deviations < 0)
test_statsitic <- min(dev_neg, nrow(males)-dev_neg)
2*pbinom(test_statsitic, nrow(males), 0.5)
[1] 1
```

For Females

```
dev_neg <- sum(females$deviations < 0)
test_statsitic <- min(dev_neg, nrow(females)-dev_neg)
2*pbinom(test_statsitic, nrow(females), 0.5)
[1] 1
```

#Since the p values for both the tests are greater than 5%, there is no significant evidence to reject the null hypothesis that the deviations follow Binomial (31,0.5) distribution for both genders.

(3)
[37 Marks]

Q. 3)

```

#### installing packages (as required)
#install.packages('dplyr')
#install.packages('ggplot2')

#### loading Packages
library(dplyr)
library(ggplot2)

#### 1.

# setting the seed
set.seed(100)
n<-100

# Generating the data for each sport
type1=matrix(c(1:n,sample(190:204,n,replace=TRUE),
               sample(90:105,n,replace=TRUE),rep(1,n)),ncol=4)
type1<-as.data.frame(type1) %>% mutate()
type2=matrix(c((n+1):(2*n),sample(178:196,n,replace=TRUE),
               sample(84:98,n,replace=TRUE),rep(2,n)),ncol=4)
type2<-as.data.frame(type2)
type3=matrix(c((2*n+1):(3*n),sample(174:191,n,replace=TRUE),
               sample(98:124,n,replace=TRUE),rep(3,n)),ncol=4)
type3<-as.data.frame(type3)

# combining the data
data <- rbind(type1,type2,type3) %>%
  mutate(Sport = case_when(V4==1~"Basketball",
                           V4==2~"Baseball",
                           V4==3~"Football")) %>%
  select(Height = V2, Weight = V3, Sport)

# printing the first five records of each sport
print(data[c(1:5,101:105,201:205),])

```

```

  Height Weight  Sport
1    199   101 Basketball
2    196    93 Basketball
3    195    94 Basketball
4    192    98 Basketball
5    198    99 Basketball
101   181    85  Baseball
102   179    90  Baseball
103   188    90  Baseball
104   191    92  Baseball
105   192    98  Baseball
201   186   112  Football

```

```
202 190 112 Football
203 185 109 Football
204 188 117 Football
205 183 106 Football
```

(5)

```
### ii.
```

```
summary(data)
```

```
      Height      Weight      Sport
Min. :174.0 Min. : 84.00 Length:300
1st Qu.:183.0 1st Qu.: 92.00 Class :character
Median :190.0 Median : 98.00 Mode  :character
Mean  :189.1 Mean   : 99.74
3rd Qu.:195.0 3rd Qu.:105.00
Max.   :204.0 Max.   :124.00
```

```
summary(data %>% filter(Sport=="Basketball"))
```

```
      Height      Weight      Sport
Min. :190.0 Min. : 90.00 Length:100
1st Qu.:193.0 1st Qu.: 94.75 Class :character
Median :196.5 Median : 98.50 Mode  :character
Mean  :197.1 Mean   : 97.89
3rd Qu.:201.0 3rd Qu.:101.00
Max.   :204.0 Max.   :105.00
```

```
summary(data %>% filter(Sport=="Baseball"))
```

```
      Height      Weight      Sport
Min. :178.0 Min. :84.00 Length:100
1st Qu.:184.0 1st Qu.:86.75 Class :character
Median :188.0 Median :90.00 Mode  :character
Mean  :187.8 Mean   :90.28
3rd Qu.:191.0 3rd Qu.:94.00
Max.   :196.0 Max.   :98.00
```

```
summary(data %>% filter(Sport=="Football"))
```

```
      Height      Weight      Sport
Min. :174.0 Min. : 98 Length:100
1st Qu.:179.0 1st Qu.:105 Class :character
Median :183.0 Median :111 Mode  :character
Mean  :182.5 Mean   :111
3rd Qu.:186.2 3rd Qu.:117
Max.   :191.0 Max.   :124
```

```
# The heights of Basketball players are quite high compared to the other two sports players
# The weights of Football players are higher than other sport players
```

```
# Baseball players have lower weight and average heights
# It is quite possible to predict the players sport based on the height and weight
# since there are significant differences observed.
```

(3)

```
### iii.
```

```
## a.
decision_tree=function(x,y){
  ifelse(y>105,"Football",
    ifelse(x>196,"Basketball","Baseball"))}

```

```
## b.
decision_tree(190,100)
[1] "Baseball"
```

```
## c.
data <- data %>%
  mutate(Decision_tree = decision_tree(Height, Weight))

```

```
## d.
table(data$Sport,data$Decision_tree)
```

	Baseball	Basketball	Football
Baseball	100	0	0
Basketball	50	50	0
Football	28	0	72

```
## e.
Basketball_gi = 1 - (50/50)^2
Basketball_gi
[1] 0
```

```
Baseball_gi = 1 - (100/178)^2 - (50/178)^2 - (28/178)^2
Baseball_gi
[1] 0.5807348
```

```
Football_gi = 1 - (72/72)^2
Football_gi
[1] 0
```

```
## f.
# Gini index for Basketball and Football are 0, indicating prediction of this sport is reliable
# Gini index for Baseball is quite high, indicating a lot of players get misclassified as
Baseball players.
```

(9)

```
### iv.
```

```

## a.
# There are b -a +1 values that the variable can take. Each one is equally likely
# The likelihood of  $k < a$  and  $k > b$  is 0
# The likelihood of  $a \leq k \leq b$  is  $1/(b-a+1)$ 

## b.
likelihood_basketball=function(x,y){
  return(ifelse(x<190 | x>204,0, 1/(204-190+1))*
    ifelse(y<90 | y>105,0, 1/(105-90+1)))
}
likelihood_baseball=function(x,y){
  return(ifelse(x<178 | x>196,0, 1/(196-178+1))*
    ifelse(y<84 | y>98,0, 1/(98-84+1)))
}
likelihood_football=function(x,y){
  return(ifelse(x<174 | x>191,0, 1/(191-174+1))*
    ifelse(y<98 | y>124,0, 1/(124-98+1)))
}
## c.
naive_bayes <- function(x,y){
  basketball <- 0.3*likelihood_basketball(x,y)
  baseball <- 0.4*likelihood_baseball(x,y)
  football <- 0.3*likelihood_football(x,y)
  if(basketball==max(basketball,baseball,football))
    return("Basketball")
  if(baseball==max(basketball,baseball,football))
    return("Baseball")
  if(football==max(basketball,baseball,football))
    return("Football")
}

## d.
naive_bayes(190,100)
[1] "Basketball"

## e.
for (i in 1:nrow(data)) {
  data$naive_Bayes[i] <- naive_bayes(data$Height[i],data$Weight[i])
}

## f.
table(data$Sport,data$naive_Bayes)

      Baseball Basketball Football
Baseball      100         0         0
Basketball     30        70         0
Football        4         4        92

## g.
Basketball_gi = 1 - (70/74)^2 - (4/74)^2
Basketball_gi

```

```
[1] 0.1022644
```

```
Baseball_gi = 1 - (100/134)^2 - (30/134)^2 - (4/134)^2
```

```
Baseball_gi
```

```
[1] 0.3920695
```

```
Football_gi = 1 - (92/92)^2
```

```
Football_gi
```

```
[1] 0
```

```
## h.
```

```
# Gini index for Football is 0, indicating prediction of this sport is reliable
```

```
# Gini index for Basketball is low but Baseball is high.
```

```
# It is comparatively better than the decision tree.
```

(13)

[30 Marks]
